



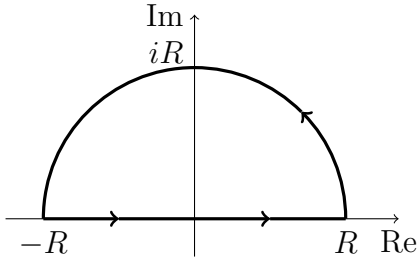
3 Complex Analysis 2024 Tutorial 7

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Cauchy's integral formula

Problem 28 Consider the path γ indicated on the figure below, with $R > 1$:



We denote by γ_1 the segment $[-R, R]$, and by γ_2 the arc.

- (a) Prove that $\int_{\gamma} \frac{1}{1+z^2} dz = \pi$ and that $\lim_{R \rightarrow \infty} \left| \int_{\gamma_2} \frac{1}{1+z^2} dz \right| = 0$.
- (b) Conclude that $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{1+x^2} dx = \pi$.

Solution.

- (a) Notice that $1+z^2=0$ for $z=\pm i$. Only the point i lies inside of our path.

We are going to apply Cauchy's Integral Formula for the path γ and the term $\frac{1/(z+i)}{z-i}$. As we haven't consider Cauchy's Integral Formulas for γ not being a circle (there are such formulas), we need to bring a circle into the picture somehow, for example we consider the three paths η_1, η_2, η_3 as in the picture below.

By Proposition 2.4,

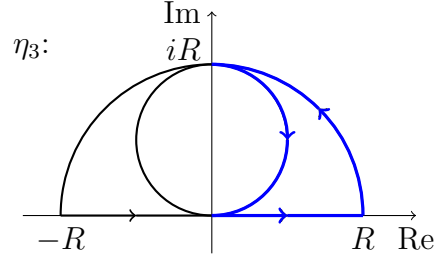
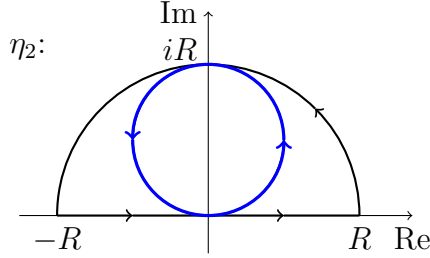
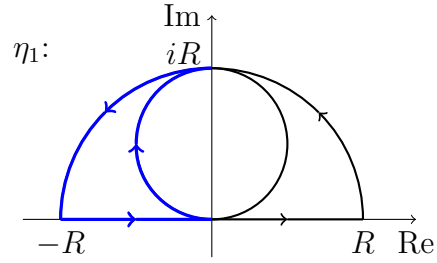
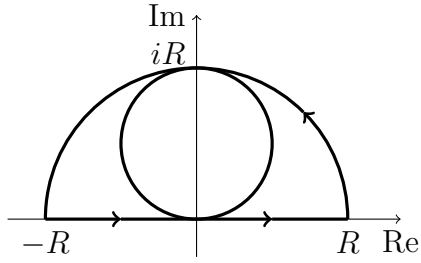
$$\int_{\gamma} \frac{1}{1+z^2} dz = \int_{\eta_1} \frac{1}{1+z^2} dz + \int_{\eta_2} \frac{1}{1+z^2} dz + \int_{\eta_3} \frac{1}{1+z^2} dz.$$

By Cauchy's Integral Theorem,

$$\int_{\eta_1} \frac{1}{1+z^2} dz = \int_{\eta_3} \frac{1}{1+z^2} dz = 0.$$

Thus,

$$\int_{\gamma} \frac{1}{1+z^2} dz = \int_{\eta_2} \frac{1/(z+i)}{z-i} dz = 2\pi i \frac{1}{i+i} = \pi.$$



To show that $\lim_{R \rightarrow \infty} \left| \int_{\gamma_2} \frac{1}{1+z^2} dz \right| = 0$, we apply the Estimation Theorem on the arc γ_2 . On the arc we have $|z| = R$, and hence $|1+z^2| \geq |z|^2 - 1 = R^2 - 1$, so that $\left| \frac{1}{1+z^2} \right| \leq \frac{1}{R^2-1}$. The length of the arc is πR . Thus,

$$\left| \int_{\gamma_2} \frac{1}{1+z^2} dz \right| \leq \frac{\pi R}{R^2-1} \rightarrow 0, \quad R \rightarrow \infty.$$

(b) We have that

$$\int_{\gamma} \frac{1}{1+z^2} dz = \int_{\gamma_1} \frac{1}{1+z^2} dz + \int_{\gamma_2} \frac{1}{1+z^2} dz = \pi.$$

It follows that

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{1+x^2} dx = \pi - \lim_{R \rightarrow \infty} \int_{\gamma_2} \frac{1}{1+z^2} dz = \pi.$$

Problem 29 Evaluate the following integrals.

(a) $\int_{\gamma} \frac{z^7 + 1}{z^2(z^4 + 1)} dz$, where $\gamma(t) = 2 + e^{it}$, $t \in [0, 2\pi]$.

(b) $\int_{\gamma} \frac{e^{-z}}{(z+2)^3} dz$, where $\gamma(t) = 3e^{it}$, $t \in [0, 2\pi]$.

(c) $\int_{\gamma} \frac{1}{(z-1)(z-2)^2} dz$, where $\gamma(t) = 2 + \frac{1}{2}e^{it}$, $t \in [0, 2\pi]$.

(d) $\int_{\gamma} \frac{z^4 e^z}{(z-1)^4} dz$, where $\gamma(t) = 2e^{it}$, $t \in [0, 2\pi]$.

Solution.

- (a) The zeros of the denominator $z^2(z^4 + 1)$ are 0 and the four fourth roots of i — they can be given explicitly, but we just notice that the solutions of $z^4 + 1 = 0$ have $|z| = 1$, and $z = 1$ is not a solution. Therefore, all zeros of the denominator $z^2(z^4 + 1)$ lie outside of the circle γ with center 2 and radius 1. It follows that

$$\int_{\gamma} \frac{z^7 + 1}{z^2(z^4 + 1)} dz = 0.$$

- (b) By Cauchy's Integral Formula for derivatives,

$$\int_{\gamma} \frac{e^{-z}}{(z + 2)^3} dz = \frac{2\pi i}{2!} (e^{-z})'' \Big|_{z=-2} = \pi i e^{-2}.$$

- (c) Notice that the point $z = 1$ lies outside of the circle γ . Thus,

$$\int_{\gamma} \frac{1}{(z - 1)(z - 2)^2} dz = \frac{2\pi i}{1!} \left(\frac{1}{z - 1} \right)' \Big|_{z=2} = 2\pi i \frac{-1}{(z - 1)^2} \Big|_{z=2} = -2\pi i.$$

- (d) Here we have

$$\int_{\gamma} \frac{z^4 e^z}{(z - 1)^4} dz = \frac{2\pi i}{3!} (z^4 e^z)^{(3)} \Big|_{z=1} = \frac{\pi i}{3} ((24z + 36z^2 + 12z^3 + z^4)e^z) \Big|_{z=1} = \frac{73\pi i e}{3}.$$

Problem 30 Let G be a region, $z_0 \in G$, $r > 0$ such that the closed disc $\overline{D}(z_0; r) \subseteq G$. Denote by γ the circle with center z_0 , radius r and positive orientation. Let f be a holomorphic function in G . Calculate

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(w) - f(z)}{(w - z)^2} dw$$

for $z \in D(z_0; r)$.

Solution.

We apply Cauchy's Integral Formula for derivatives to the function $w \mapsto f(w) - f(z)$ with fixed $z \in D(z_0; r)$. Its derivative is $f'(w)$, and for $w = z$ it takes the value $f'(z)$. We obtain

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(w) - f(z)}{(w - z)^2} dw = f'(z).$$

Problem 31 Let f be a holomorphic function in the disc $D(0; 1) = \{z \in \mathbb{C} : |z| < 1\}$ such that $|f(z)| \leq 1$ for all $z \in D(0; 1)$. Show that

$$|f'(z)| \leq 4 \quad \text{for } |z| \leq \frac{1}{2}.$$

Solution.

Let γ be a circle with center 0 and radius r , where $\frac{1}{2} < r < 1$. Let $|z| \leq \frac{1}{2}$. By Cauchy's Integral Formula for derivatives,

$$f'(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w-z)^2} dw.$$

By the Estimation Theorem,

$$|f'(z)| \leq \frac{1}{2\pi} \max_{|w|=r} \left| \frac{f(w)}{(w-z)^2} \right| \cdot 2\pi r.$$

By assumption, $|f(w)| \leq 1$. If $|w| = r > \frac{1}{2}$ and $|z| \leq \frac{1}{2}$, then $|w-z| \geq |w| - |z| \geq r - \frac{1}{2}$. Thus,

$$|f'(z)| \leq \frac{r}{\left(r - \frac{1}{2}\right)^2}.$$

Since this is true for any $\frac{1}{2} < r < 1$, we have

$$|f'(z)| \leq \frac{1}{\left(1 - \frac{1}{2}\right)^2} = 4.$$